

Class Diagram

Project: LearnLens AI

1. System Class Structure & Domain Model

```
classDiagram
direction TB

class User {
+String id
+String name
+String email
+startSession(title, platform)
+pauseAgent()
+resumeAgent()
+issueCommand(cmd)
}

class LearningSession {
+String id
+String title
+String source_platform
+String source_url_or_title
+Float created_at
+Float updated_at
+Float duration_sec
+Int progress_pct
+TaskState status
+Boolean is_pinned
+List~String~ tags
}

class ChromeSource {
+String streamId
+String displaySurface
+Boolean audioTrackPresent
+getMediaStream()
+stopStream()
}

class CaptureManager {
+Boolean isCapturing
+Float sampleIntervalSec
+startDisplayCapture(options)
+stopDisplayCapture()
+grabFrame()
}

class AudioCapture {
+Float sampleRate
+Int channels
+bufferAudio(chunk)
+exportWav()
}

class FrameCapture {
+String id
+String session_id
```

```

+Float timestamp_sec
+String timestamp_formatted
+String image_path
+String thumbnail_path
+String p_hash
+String ocr_text
+String visual_description
+VisualCategory category
+Float importance_score
+List~String~ concepts
+Boolean is_pinned
}

class TranscriptSegment {
+String id
+String session_id
+Float timestamp_start
+Float timestamp_end
+String timestamp_formatted
+String speaker
+String text
+Float confidence
+String topic
+String associated_frame_id
}

class Concept {
+String id
+String session_id
+String name
+String definition
+String simple_explanation
+String deep_explanation
+String evidence_timestamp
+String evidence_frame_id
+List~String~ related_concepts
+String exam_importance
+String difficulty
+Float confidence
+Boolean is_pinned
}

class Embedding {
+List~Float~ vector
+Int dimension
+cosineSimilarity(other)
}

class VectorStore {
+Map~String, List~Float~~ index
+insert(id, vector)
+queryTopK(queryVec, k)
}

class RAGRetriever {
+retrieveContext(sessionId, query, topK)
+assemblePromptContext(results)
}

class LearnerMemory {
+String session_id
+List~String~ concepts_viewed
}

```

```

+List-String~ concepts_grasped
+List-String~ concepts_struggling
+Int questions_asked_count
+Float practice_accuracy
+List-String~ weak_areas
+recordQuestion(q)
+recordQuizAttempt(isCorrect)
}

class ChatMessage {
+String id
+String session_id
+String sender
+String text
+TeacherMode mode
+Float timestamp
+List~EvidenceItem~ evidence
+Boolean is_pinned
}

class AITeacher {
+teach(sessionId, message, mode)
+explainConcept(conceptId)
+remedialLesson(weakArea)
}

class TaskManager {
+Map~String, TaskState~ taskStates
+pauseTask(sessionId)
+resumeTask(sessionId)
+executeCommand(sessionId, cmd)
}

class PPTGenerator {
+generatePresentation(sessionId, style, count)
+buildTitleSlide(prs, title)
+embedDiagramSlide(prs, frame)
}

class PDFGenerator {
+generateVisualPack(sessionId)
+generateTeachingReport(sessionId)
}

class QuizImporter {
+importPdf(sessionId, pdfPath)
+parseQuestions(rawText)
+mapConcepts(questions)
}

class QuizQuestion {
+String id
+String session_id
+String question
+List-String~ options
+Int correct_option_index
+String concept_tested
+String relevant_timestamp
+String explanation
+Int user_answer_index
}

```

```

%% Provider Abstractions
class ModelProvider {
<<interface>>
+is_available() Boolean
}

class LLMPProvider {
<<interface>>
+generate_response(prompt, system_prompt)
}

class VLMPProvider {
<<interface>>
+analyze_image(image_path, prompt)
}

class TranscriptionProvider {
<<interface>>
+transcribe_audio_file(audio_path)
}

class OCRProvider {
<<interface>>
+extract_text(image_path)
}

%% Relationships
User "1" --> "*" LearningSession : owns
LearningSession "1" *-- "1" ChromeSource : authorized from
LearningSession "1" *-- "*" FrameCapture : contains
LearningSession "1" *-- "*" TranscriptSegment : contains
LearningSession "1" *-- "*" Concept : contains
LearningSession "1" *-- "*" ChatMessage : records
LearningSession "1" *-- "1" LearnerMemory : maintains
LearningSession "1" *-- "*" QuizQuestion : tracks

CaptureManager --> ChromeSource : captures
CaptureManager --> FrameCapture : produces
CaptureManager --> AudioCapture : routes audio

AudioCapture --> TranscriptionProvider : passes audio
FrameCapture --> OCRProvider : extracts text
FrameCapture --> VLMPProvider : classifies

Concept --> Embedding : embedded in
VectorStore --> Embedding : stores
RAGRetriever --> VectorStore : queries
RAGRetriever --> LearningSession : scopes to

AITeacher --> RAGRetriever : retrieves grounding
AITeacher --> LLMPProvider : prompts
AITeacher --> LearnerMemory : updates

PPTGenerator --> LearningSession : extracts insights
PPTGenerator --> FrameCapture : embeds diagrams
PDFGenerator --> LearningSession : formats
QuizImporter --> QuizQuestion : produces
TaskManager --> LearningSession : governs state

ModelProvider <|-- LLMPProvider
ModelProvider <|-- VLMPProvider
ModelProvider <|-- TranscriptionProvider

```

ModelProvider <|-- OCRProvider